

Kexin LIN

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Education

Université Claude Bernard Lyon 1 <i>PhD student, funded by ERC Advanced Grant EYAWKAJKOS</i> • Supervisor: Filippo Santambrogio • Thesis (on going): Higher-order gradient flows in the Wasserstein space	2024.9 – Present <i>Lyon, France</i>
École Normale Supérieure de Lyon <i>Master 2 in PDE, funded by Labex MILYON</i> • Result: 17.6/20 • Internship: Existence of the TV Wasserstein Gradient Flow, supervised by Filippo Santambrogio	2023.9 – 2024.9 <i>Lyon, France</i>
Beihang University <i>Undergraduate student at Hua Loo-Keng Experimental Class for Mathematics</i> • GPA: 3.87/4.0 • Undergraduate thesis: Elementary Geometric Regularity of Sets and Functions in Euclidean Spaces, supervised by Xiangyu Liang	2019.9 – 2023.6 <i>Beijing, China</i>

Papers

- L., Santambrogio, F. (2026). Existence of a solution of the TV Wasserstein gradient flow. arXiv preprint arXiv:2601.22847.

Talks, Posters

Seizième journée annuelle de l'équipe MMCS de l'ICJ <i>École Centrale de Lyon</i> • Talk: The dissipation rate of internal functionals along the JKO scheme	2026.7 <i>Lyon, France</i>
Optimal Transport in GMT, Geometric Measure Theory in OT <i>Dipartimento di Matematica "Tullio Levi-Civita"</i> • Talk: Existence of a solution of the TV Wasserstein gradient flow	2026.3 <i>Padova, Italy</i>
Calculus of Variations and applications <i>CIRM</i> • Talk: Decay in the JKO Scheme for the Total Variation Flow in Wasserstein Space	2026.1 <i>Luminy, France</i>
Gradient Flow Face to Face 5 <i>Institute of Mathematics of the University of Granada</i> • Short talk: Existence of the TV Wasserstein Gradient Flow	2025.9 <i>Granada, Spain</i>
Geometric Methods in Calculus of Variations <i>Centro di Ricerca Matematica Ennio De Giorgi</i> • Poster: Existence of the TV Wasserstein Gradient Flow via JKO scheme	2025.6 <i>Pisa, Italy</i>
GT EYAWKAJKOS <i>Institut Camille Jordan, Université Claude Bernard Lyon 1</i> • Reading seminar: On the total variation Wasserstein gradient flow and the TV-JKO scheme	2024.5 <i>Lyon, France</i>

Research stays

Dipartimento di Matematica "Tullio Levi-Civita" - Padova - Two weeks	2026.3
<i>Funded by Arqus</i>	

Teaching Experience

Teaching Assistant of <i>Real Analysis</i>	2022.2 - 2023.6
<i>School of Mathematics and System Science, Beihang University</i>	<i>Beijing, China</i>
• Works involving giving tutorial sessions and grading homework	
Teaching Assistant of <i>Mathematical Analysis</i>	2022.9 - 2023.6
<i>School of Mathematics and System Science, Beihang University</i>	<i>Beijing, China</i>
• Works involving giving tutorial sessions and grading homework	

Language

Chinese: Native speaker
English: Fluent
French: Intermediate